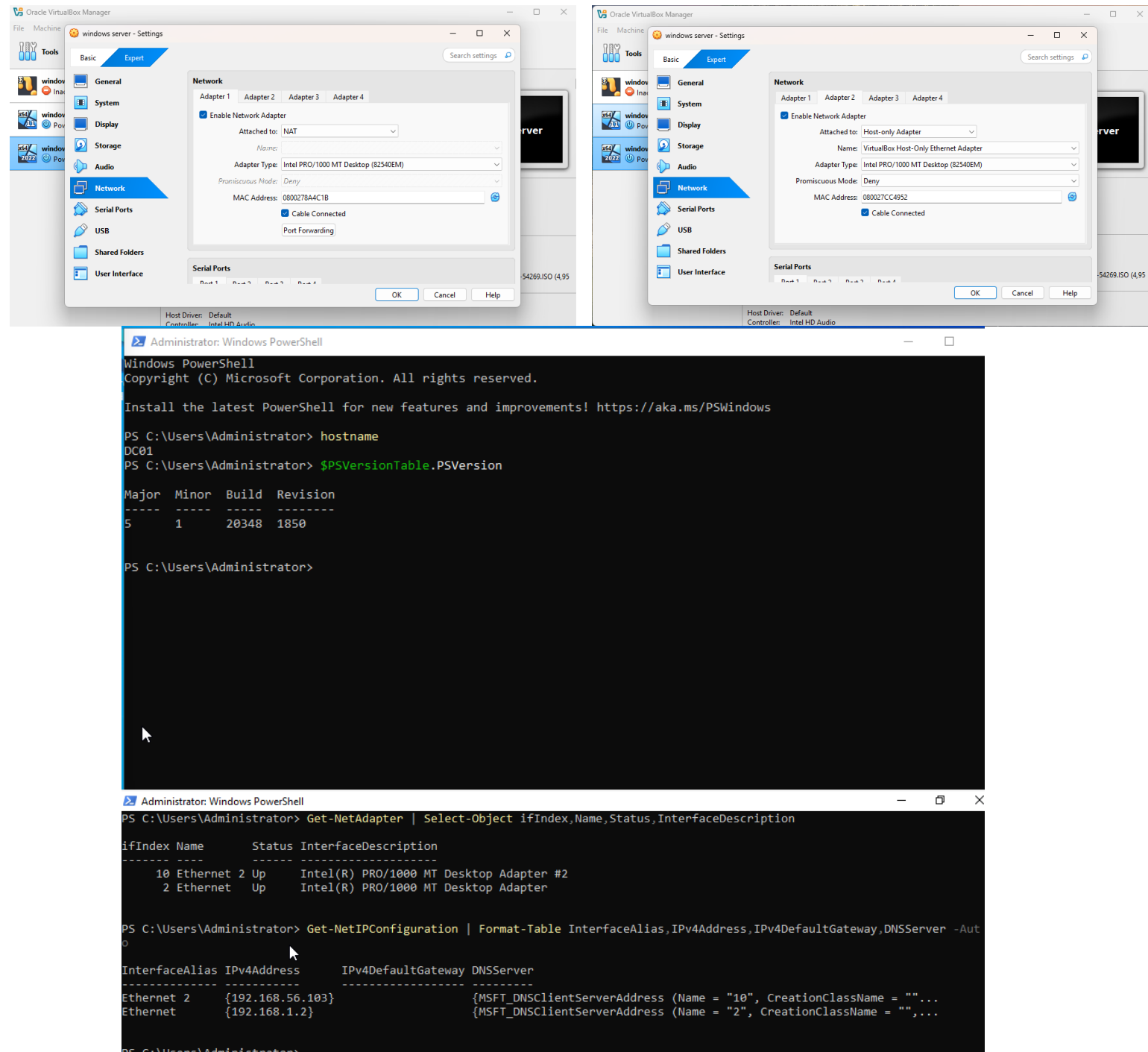


IPv4 IPv6

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Windows Administration
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Initial Network Configuration and Environment Setup



Before configuring IPv4 and IPv6 on Windows Server, both network adapters were prepared in Oracle VirtualBox. Adapter 1 was attached to NAT to provide Internet access, while Adapter 2 was set to Host-only Adapter to create an isolated local network for laboratory experiments.

In PowerShell, the commands *hostname* and *\$PSVersionTable.PSVersion* were executed to confirm the server identity and PowerShell version.

The command *Get-NetAdapter* displayed the available interfaces and their operational status, while *Get-NetIPConfiguration* showed each adapter's current IPv4 address, gateway, and DNS configuration. These preliminary checks ensured that both adapters were correctly detected and ready for further IPv4 and IPv6 configuration.

Troubleshooting DHCP and DNS Connectivity Issues

```
PS C:\Users\Administrator> ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : domenis.ut.ee
    IPv6 Address. . . . . : fd17:625c:f037:2:61b4:b5ff:db4e:a5bf
    Link-local IPv6 Address . . . . . : fe80::f958:a980:45cd:136d%2
    IPv4 Address. . . . . : 192.168.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%2

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::684f:b541:8c02:4605%10
    Default Gateway . . . . . :

PS C:\Users\Administrator> ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : domenis.ut.ee
    IPv6 Address. . . . . : fd17:625c:f037:2:61b4:b5ff:db4e:a5bf
    Link-local IPv6 Address . . . . . : fe80::f958:a980:45cd:136d%2
    IPv4 Address. . . . . : 192.168.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%2

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::684f:b541:8c02:4605%10
    IPv4 Address. . . . . : 192.168.56.103
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

```
PS C:\Users\Administrator> Resolve-DnsName microsoft.com -Type A
Resolve-DnsName : microsoft.com : DNS server failure
At line:1 char:1
+ Resolve-DnsName microsoft.com -Type A
+ ~~~~~
+ CategoryInfo          : ResourceUnavailable: (microsoft.com:String) [Resolve-DnsName], Win32Exception
+ FullyQualifiedErrorId : RCODE_SERVER_FAILURE,Microsoft.DnsClient.Commands.ResolveDnsName

PS C:\Users\Administrator> Test-NetConnection microsoft.com -Port 443 -InformationLevel Detailed
WARNING: Name resolution of microsoft.com failed

ComputerName           : microsoft.com
RemoteAddress          :
NameResolutionResults  :
InterfaceAlias         :
SourceAddress          :
NetRoute (NextHop)    :
PingSucceeded          : False

Administrator: Windows PowerShell

PS C:\Users\Administrator> Disable-NetAdapter -Name "Ethernet 2" -Confirm:$false
PS C:\Users\Administrator> netsh int ip reset
Resetting Compartment Forwarding, OK!
Resetting Compartment, OK!
Resetting Control Protocol, OK!
Resetting Echo Sequence Request, OK!
Resetting Global, OK!
Resetting Interface, OK!
Resetting Anycast Address, OK!
Resetting Multicast Address, OK!
Resetting Unicast Address, OK!
Resetting Neighbor, OK!
Resetting Path, OK!
Resetting Potential, OK!
Resetting Prefix Policy, OK!
Resetting Proxy Neighbor, OK!
Resetting Route, OK!
Resetting Site Prefix, OK!
Resetting Subinterface, OK!
Resetting Wakeup Pattern, OK!
Resetting Resolve Neighbor, OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , failed.
Access is denied.

Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Resetting , OK!
Restart the computer to complete this action.

PS C:\Users\Administrator> netsh winsock reset

Sucessfully reset the Winsock Catalog.
You must restart the computer in order to complete the reset.
```

```

PS C:\Users\Administrator> ipconfig /release "Ethernet"

Windows IP Configuration

Adapter Ethernet is not enabled for DHCP.
PS C:\Users\Administrator> ipconfig /renew "Ethernet"

Windows IP Configuration

Adapter Ethernet is not enabled for DHCP.
PS C:\Users\Administrator> Set-NetIPInterface -InterfaceAlias "Ethernet" -Dhcp Enabled
PS C:\Users\Administrator> ipconfig /release "Ethernet"

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : domenis.ut.ee
    IPv6 Address. . . . . : fd17:625c:f037:2:61b4:b5ff:db4e:a5bf
    Link-local IPv6 Address . . . . . : fe80::f958:a980:45cd:136d%2
    Default Gateway . . . . . : fe80::2%2
PS C:\Users\Administrator> ipconfig /renew "Ethernet"

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : domenis.ut.ee
    IPv6 Address. . . . . : fd17:625c:f037:2:61b4:b5ff:db4e:a5bf
    Link-local IPv6 Address . . . . . : fe80::f958:a980:45cd:136d%2
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%2
                             10.0.2.2
PS C:\Users\Administrator> Get-NetIPConfiguration -InterfaceAlias "Ethernet" | Format-List IPv4Address,IPv4DefaultGateway,DNSServer
IPv4Address : {10.0.2.15}
DNSServer   : {MSFT DNSClientServerAddress (N Task View ", CreationClassName = "", SystemCreationClassName = "", SystemName =

```

```

PS C:\Users\Administrator> Enable-NetAdapter -Name "Ethernet 2"
PS C:\Users\Administrator> Get-NetAdapter | Select-Object Name, Status

Name      Status
----      -
Ethernet 2 Up
Ethernet  Up

```

During the configuration of IPv4 connectivity, the command ***ipconfig /release*** and ***ipconfig /renew*** were used to refresh the DHCP-assigned addresses. However, name resolution failed, as shown by the error “**DNS server failure**” when executing ***Resolve-DnsName microsoft.com***. This indicated a problem with the DNS client configuration on the NAT adapter.

To address this, the Host-only adapter was temporarily disabled using ***Disable-NetAdapter***, and the commands ***netsh int ip reset*** and ***netsh winsock reset*** were executed to reset the TCP/IP and Winsock stacks.

Further diagnostics revealed that the NAT adapter was not configured for DHCP, producing the error “*Adapter Ethernet is not enabled for DHCP.*” The issue was resolved with the command ***Set-NetIPInterface -InterfaceAlias "Ethernet" -Dhcp Enabled***, after which ***ipconfig /renew*** successfully obtained a new IP address, gateway, and DNS server.

Finally, ***Enable-NetAdapter*** was executed to re-enable the Host-only adapter, restoring both interfaces to the “Up” status for further network configuration.

Configuring a Static IPv4 Address for the Host-Only Network

```
PS C:\Users\Administrator> Get-NetIPAddress -InterfaceAlias "Ethernet 2" -AddressFamily IPv4 | Remove-NetIPAddress -Confirm:$false
PS C:\Users\Administrator> New-NetIPAddress -InterfaceAlias "Ethernet 2" -IPAddress 192.168.56.10 -PrefixLength 24

IPAddress           : 192.168.56.10
InterfaceIndex      : 10
InterfaceAlias       : Ethernet 2
AddressFamily        : IPv4
Type                 : Unicast
PrefixLength         : 24
PrefixOrigin         : Manual
SuffixOrigin         : Manual
AddressState         : Tentative
ValidLifetime        : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime    : Infinite ([TimeSpan]::MaxValue)
SkipAsSource         : False
PolicyStore          : ActiveStore

IPAddress           : 192.168.56.10
InterfaceIndex      : 10
InterfaceAlias       : Ethernet 2
AddressFamily        : IPv4
Type                 : Unicast
PrefixLength         : 24
PrefixOrigin         : Manual
SuffixOrigin         : Manual
AddressState         : Invalid
ValidLifetime        : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime    : Infinite ([TimeSpan]::MaxValue)
SkipAsSource         : False
PolicyStore          : PersistentStore

PS C:\Users\Administrator> ping 192.168.56.10

Pinging 192.168.56.10 with 32 bytes of data:
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128
Reply from 192.168.56.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.56.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PS C:\Users\Administrator> Get-NetIPConfiguration -InterfaceAlias "Ethernet 2"

InterfaceAlias       : Ethernet 2
InterfaceIndex       : 10
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter #2
NetProfile.Name       : lab.local
IPv4Address           : 192.168.56.10
IPv6DefaultGateway   :
IPv4DefaultGateway    :
DNSServer             : fec0:0:0:ffff::1
                     : fec0:0:0:ffff::2
                     : fec0:0:0:ffff::3
```

To establish a local network connection between virtual machines, a static IPv4 address was configured on the Host-Only adapter (Ethernet 2). First, any existing IPv4 addresses were removed using **Get-NetIPAddress | Remove-NetIPAddress**, ensuring a clean configuration.

The command **New-NetIPAddress -InterfaceAlias "Ethernet 2" -IPAddress 192.168.56.10 -PrefixLength 24** was then executed to assign a permanent address within the Host-Only subnet. The configuration was verified using **Get-NetIPConfiguration**, which confirmed that the adapter was successfully assigned the address 192.168.56.10/24.

Finally, connectivity was tested with the **ping 192.168.56.10** command, which returned 100% successful replies, indicating that the Host-Only interface was fully functional and ready for further IPv6 configuration.

IPv6 Configuration, Routing Tables, and Connectivity Verification

After configuring IPv4, the focus shifted to the IPv6 stack. The **route print** command displayed both IPv4 and IPv6 routing tables, confirming that the default IPv4 gateway was 10.0.2.2 (NAT) and that the Host-Only adapter had a local IPv6 prefix 2001:db8:56:1::/64.

```
PS C:\Users\Administrator> route print
=====
Interface List
2...08 00 27 8a 4c 1b .....Intel(R) PRO/1000 MT Desktop Adapter
10...08 00 27 cc 49 52 .....Intel(R) PRO/1000 MT Desktop Adapter #2
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
0.0.0.0                    0.0.0.0          10.0.2.2          10.0.2.15        25
10.0.2.0                   255.255.255.0    On-link           10.0.2.15        281
10.0.2.15                  255.255.255.255  On-link           10.0.2.15        281
10.0.2.255                 255.255.255.255  On-link           10.0.2.15        281
127.0.0.0                  255.0.0.0        On-link           127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link           127.0.0.1        331
127.255.255.255            255.255.255.255  On-link           127.0.0.1        331
192.168.56.0               255.255.255.0    On-link           192.168.56.10     281
192.168.56.10              255.255.255.255  On-link           192.168.56.10     281
192.168.56.255             255.255.255.255  On-link           192.168.56.10     281
224.0.0.0                  240.0.0.0        On-link           127.0.0.1        331
224.0.0.0                  240.0.0.0        On-link           10.0.2.15        281
224.0.0.0                  240.0.0.0        On-link           192.168.56.10     281
255.255.255.255            255.255.255.255  On-link           127.0.0.1        331
255.255.255.255            255.255.255.255  On-link           10.0.2.15        281
255.255.255.255            255.255.255.255  On-link           192.168.56.10     281
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
2 281 ::/0 fe80::2
1 331 ::1/128 On-link
10 281 2001:db8:56:1::/64 On-link
10 281 2001:db8:56:1::10/128 On-link
2 281 fd17:625c:f037:2::/64 On-link
2 281 fd17:625c:f037:2::61b4:b5ff:dbda:a5bf/128 On-link
2 281 fe80::/64 On-link

2 281 fe80::f958:a980:45cd:136d/128 On-link
1 331 ff00::/8 On-link
2 281 ff00::/8 On-link
10 281 ff00::/8 On-link
=====
Persistent Routes:
None
```

The command ***Get-NetAdapterBinding*** verified that the Internet Protocol Version 6 (TCP/IPv6) component was enabled on the Host-Only adapter. Once IPv6 was active, ***Resolve-DnsName microsoft.com -Type AAAA*** successfully retrieved multiple IPv6 (AAAA) records, demonstrating proper DNS resolution for IPv6 hosts.

```
PS C:\Users\Administrator> Get-NetAdapterBinding -Name "Ethernet 2" | Where-Object ComponentID -eq "ms_tcpip6"
```

Name	DisplayName	ComponentID	Enabled
Ethernet 2	Internet Protocol Version 6 (TCP/IPv6)	ms_tcpip6	True

```
PS C:\Users\Administrator> Resolve-DnsName microsoft.com -Type AAAA
```

Name	Type	TTL	Section	IPAddress
microsoft.com	AAAA	2678	Answer	2603:1020:201:10::10f
microsoft.com	AAAA	2678	Answer	2603:1010:3:3::5b
microsoft.com	AAAA	2678	Answer	2603:1030:b:3::152
microsoft.com	AAAA	2678	Answer	2603:1030:20e:3::23c
microsoft.com	AAAA	2678	Answer	2603:1030:c02:8::14

Although direct ICMPv6 communication to Google's DNS server (***ping -6 2001:4860:4860::8888***) returned "Destination net unreachable," this behavior was expected because VirtualBox NAT does not forward IPv6 traffic by default.

```
PS C:\Users\Administrator> ping -6 2001:4860:4860::8888
```

Pinging 2001:4860:4860::8888 with 32 bytes of data:
Destination net unreachable.
Destination net unreachable.
Destination net unreachable.
Destination net unreachable.

Ping statistics for 2001:4860:4860::8888:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Final connectivity verification was performed using ***Test-NetConnection*** ***microsoft.com -Port 443***, which showed `TcpTestSucceeded : True`, proving that HTTPS communication was functional.

```
PS C:\Users\Administrator> Test-NetConnection microsoft.com -Port 443 -InformationLevel Detailed
```

```
PS C:\Users\Administrator> tracert -6 microsoft.com
```

Tracing route to microsoft.com [2603:1010:3:3::5b]
over a maximum of 30 hops:

1 Destination net unreachable.

Trace complete.

SourceAddress	: 10.0.2.15
NetRoute (NextHop)	: 10.0.2.2
TcpTestSucceeded	: True

The *tracert -4* and *tracert -6* commands illustrated route differences: IPv4 traffic reached the destination successfully, while IPv6 routing stopped at the local network boundary.

```
PS C:\Users\Administrator> tracert -4 microsoft.com

Tracing route to microsoft.com [13.107.213.67]
over a maximum of 30 hops:

  1      *          *          *          Request timed out.
  2      *          *          *          Request timed out.
  3      *          *          *          Request timed out.
  4      *          *          *          Request timed out.
  5      *          *          *          Request timed out.
  6      *          *          *          Request timed out.
  7      *          *          *          Request timed out.
  8      *          *          *          Request timed out.
  9      *          *          *          Request timed out.
 10     *          *          *          Request timed out.
 11     *          *          *          Request timed out.
 12     *          *          *          Request timed out.
 13     *          *          *          Request timed out.
 14     *          *          *          Request timed out.
 15     *          *          *          Request timed out.
 16     *          *          *          Request timed out.
 17     *          *          *          Request timed out.
 18     *          *          *          Request timed out.
 19     *          39 ms        *          13.107.213.67
 20     *          *          *          Request timed out.
 21     *          *          *          Request timed out.
 22     *          *          *          Request timed out.
 23     *          *          *          Request timed out.
 24     *          *          *          Request timed out.
 25     *          *          *          Request timed out.
 26     *          *          *          Request timed out.
 27     *          *          *          Request timed out.
 28     *          *          *          Request timed out.
 29     *          *          *          Request timed out.
 30     *          *          *          Request timed out.

Trace complete.
```

The *Get-NetNeighbor -AddressFamily IPv6* output listed active and unreachable neighbors, confirming that IPv6 address discovery and Neighbor Discovery Protocol (NDP) were functioning correctly within the local scope.

```
PS C:\Users\Administrator> Get-NetNeighbor -AddressFamily IPv6
```

ifIndex	IPAddress	LinkLayerAddress	State	PolicyStore
10	ff02::1:ff02:4605	33-33-FF-02-46-05	Permanent	ActiveStore
10	ff02::1:ff00:10	33-33-FF-00-00-10	Permanent	ActiveStore
10	ff02::1:3	33-33-00-01-00-03	Permanent	ActiveStore
10	ff02::1:2	33-33-00-01-00-02	Permanent	ActiveStore
10	ff02::fb	33-33-00-00-00-FB	Permanent	ActiveStore
10	ff02::16	33-33-00-00-00-16	Permanent	ActiveStore
10	ff02::c	33-33-00-00-00-0C	Permanent	ActiveStore
10	ff02::2	33-33-00-00-00-02	Permanent	ActiveStore
10	ff02::1	33-33-00-00-00-01	Permanent	ActiveStore
10	2001:db8:56:1::10	00-00-00-00-00-00	Unreachable	ActiveStore
2	ff02::1:ffcd:136d	33-33-FF-CD-13-6D	Permanent	ActiveStore
2	ff02::1:ff4e:a5bf	33-33-FF-4E-A5-BF	Permanent	ActiveStore
2	ff02::1:ff00:2	33-33-FF-00-00-02	Permanent	ActiveStore
2	ff02::1:3	33-33-00-01-00-03	Permanent	ActiveStore
2	ff02::1:2	33-33-00-01-00-02	Permanent	ActiveStore
2	ff02::fb	33-33-00-00-00-FB	Permanent	ActiveStore
2	ff02::16	33-33-00-00-00-16	Permanent	ActiveStore
2	ff02::c	33-33-00-00-00-0C	Permanent	ActiveStore
2	ff02::2	33-33-00-00-00-02	Permanent	ActiveStore
2	ff02::1	33-33-00-00-00-01	Permanent	ActiveStore
2	fe80::2	52-56-00-00-00-02	Stale	ActiveStore
1	ff02::1:2		Permanent	ActiveStore
1	ff02::16		Permanent	ActiveStore
1	ff02::c		Permanent	ActiveStore

Summary and Conclusions

This practical exercise demonstrated the configuration and verification of IPv4 and IPv6 network settings on a Windows Server virtual machine using Oracle VirtualBox. Both the NAT and Host-Only adapters were successfully configured to provide Internet access and isolated local communication.

Through PowerShell commands, static IPv4 and IPv6 addresses were assigned, DNS and DHCP services were tested, and routing tables were analyzed. Although IPv6 Internet connectivity was limited by VirtualBox's NAT restrictions, local IPv6 functionality and DNS name resolution were confirmed.

The results highlight the importance of understanding how different adapter types, IP versions, and routing mechanisms interact within a virtualized environment, ensuring proper communication and network troubleshooting in real-world system administration tasks.